



Kedist Shegute Dimore

AI & KG
Engineer



Contact



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Location

Addis Ababa, Ethiopia
UAE resident - open to relocate



GitHub

github.com/kedistS



LinkedIn

linkedin.com/in/kedist-shegute



Skills

Languages

Python · Java · C++ · JavaScript · Dart · PHP

AI & ML

TensorFlow · Keras · GNNs · MiniLM · LLM tools

Knowledge Graphs

Neo4j · BioCypher · NetworkX · MeTTa · Prolog

AI Backend

FastAPI · Docker · REST APIs · CI/CD



Profile

AI and Knowledge Graph Engineer with 3+ years of experience at iCog Labs building graph-based AI systems, biomedical knowledge graphs, neural subgraph mining workflows, LLM interpretation tools, and deployable AI product interfaces. B.Sc. Computer Science graduate from Addis Ababa University, 2024. UAE residency visa holder with a twofour54 Software Developer license, available for UAE, remote, or hybrid AI/backend roles.



Experience

Team Leader - NeuroGraph AI Assistant

iCog Labs | October 2025 to April 2026

- Led an end-to-end AI platform for graph ingestion, neural motif mining, LLM-powered motif interpretation, interactive KG visualization, and annotation workflows.
- Integrated specialized services including an AtomSpace/graph builder, neural subgraph miner, REST orchestration API, query backend, and React/Remix frontend.
- Enabled raw CSV/JSON data to become NetworkX, MeTTa, or Neo4j graph outputs with analytics dashboards, mining configuration, downloadable reports, and interpreted motif insights.

Neural Subgraph Mining & Matcher Systems

Rejuve.Bio / iCog Labs

- Worked with a GNN-based framework for neural subgraph matching and frequent subgraph mining, inspired by NeuroMatch and SPMiner research.
- Supported semantic mining workflows using typed directed graphs, cached MiniLM label embeddings, R-GCN/order-embedding models, greedy/MCTS search, and interactive HTML motif visualizations.
- Exposed mining workflows through FastAPI endpoints for job submission, status tracking, and result retrieval.

Computer Scientist - BioCypher KG & Biomedical Research

iCog Labs | April 2023 to Present

- Built and maintained biomedical KG infrastructure integrating 34 human and 11 Drosophila sources across genes, proteins, regulation, variants, diseases, ontologies, and annotations.
- Applied graph mining to a 150,619-node, ~20M-edge BioCypher KG, discovering 11,311 motif instances across recurring biological graph structures.
- Analyzed TFLink and STRING networks, reporting recurrent regulatory and protein-interaction motifs connected to biological modules and convergence patterns.



Impact

20M

edge BioCypher KG analyzed

11,311

KG pattern instances discovered

14,877

STRING PPI pattern instances analyzed

503

TFLink regulatory motifs reported



License

UAE work-ready

Residency visa holder

twofour54 Software Developer

License #3767



Languages

English

Advanced

Amharic

Native

Korean

Reading/listening



Interests

Artificial Intelligence

Graph Science

Backend Systems

Technical Writing



Tools

AI Systems

SPMiner · NeuroMatch · R-GCN · Order embeddings · Gemini

Graph & Bio

Neo4j GDS · BioCypher · TFLink · STRING · GO · UniProt



AI & Knowledge Graph Systems

Graph Mining, LLM Interpretation & Biomedical AI

- Built AI workflows that move from raw graph data to generated KG formats, neural motif mining, visual exploration, downloadable results, and LLM-based motif interpretation.
- Worked across GNN-style graph representation learning, semantic label embeddings, biological network analysis, and deployable FastAPI/Docker services for AI tooling.



Projects

Coffee Plant Disease Classification System

Final Year Project | 2023 to 2024

Built an end-to-end plant disease classification workflow for coffee leaves, including dataset collection, labeling, model training, evaluation, API integration, and frontend delivery.

Campus PC Authentication System

Personal Project | 2023

Built a Flutter mobile app for university device registration and QR-code ownership verification.

Lounge Management System

Personal Project | 2022 to 2023

Developed a Java Swing desktop app for inventory, customer ordering, and real-time order tracking.



Research

Neural Subgraph Mining Analysis of a Human Biological Knowledge Graph

Analysis Report | BioCypher KG, 2026

Analyzed a 150,619-node, ~20M-edge KG integrating GO, STRING, TFLink, GENCODE, UniProt, and GAF data; characterized 11,311 discovered instances across 18 motifs.

Hierarchical Regulatory Patterns in Human Transcription Factor Networks

Manuscript | iCog Labs / Rejuve.Bio, 2026

Analyzed the TFLink network and reported 503 recurrent hierarchical pattern instances across 374 unique genes.

Structural Motif Analysis in Protein-Protein Interaction Networks

Manuscript | iCog Labs / Rejuve.Bio, 2026

Analyzed STRING PPI mining results with 14,877 pattern instances across 18 motif types, highlighting immune, chromatin, mitochondrial, replication, and transport modules.



Education

B.Sc. in Computer Science

Addis Ababa University | Graduated: July 2024



Certifications

Coursera Specializations

- Supervised Machine Learning: Regression and Classification
- Neural Networks and Deep Learning
- Mathematics for Machine Learning: Linear Algebra
- Object-Oriented Design